

Coherence Analysis Overview

- Definition
- Properties
- Estimation
- Correlation of complex-valued RVs
- Examples
- Discussion

Definition

Coherency

$$\mathcal{G}_{xy}(\omega) \triangleq \frac{R_{xy}(e^{j\omega})}{\sqrt{R_x(e^{j\omega})R_y(e^{j\omega})}}$$

Also known as the **coherency spectrum** (Weiner, 1930) or **normalized cross-spectrum**. Similar to the correlation coefficient in frequency.

Magnitude Squared Coherence (MSC)

$$\mathcal{G}_{xy}^2(\omega) \triangleq \frac{|R_{xy}(e^{j\omega})|^2}{R_x(e^{j\omega})R_y(e^{j\omega})}$$

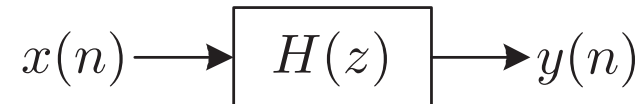
Also known as the **coherence function** and simply **coherence**

What is it Really?

$$\mathcal{G}_{xy}^2(\omega) \triangleq \frac{|R_{xy}(e^{j\omega})|^2}{R_x(e^{j\omega})R_y(e^{j\omega})}$$

- Coherence is a measure of correlation in frequency
- Usual assumptions apply (WSS)
- For estimation, we require ergodicity

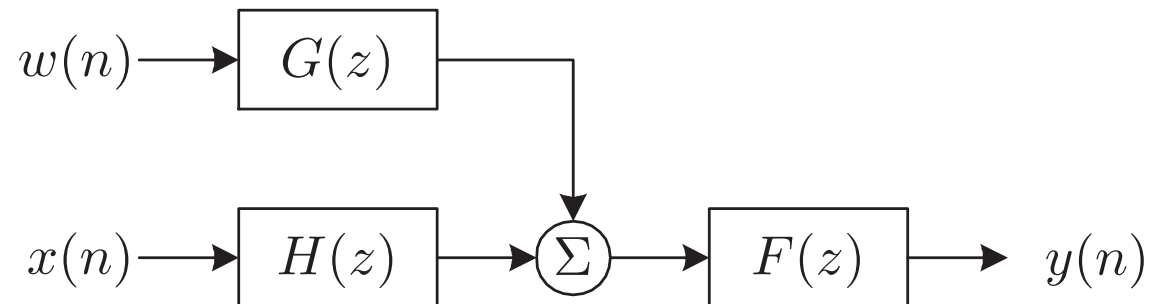
Example 1: Coherence and LTI Systems



Consider the stochastic process above where $x(n)$ and $y(n)$ are jointly wide sense stationary. Solve for the coherence $\mathcal{G}_{xy}^2(\omega)$. Hint: recall that if $y(n) = h(n) * x(n)$, then $R_{xy}(z) = H^*(z^{-*})R_x(z)$.

Example 1: Workspace

Example 2: Coherence and LTI Systems



Consider the stochastic process above where $w(n)$ and $x(n)$ are jointly wide sense stationary and uncorrelated and the LTI systems are all stable. Solve for the coherence $\mathcal{G}_{xy}^2(\omega)$.

Example 2: Workspace

Properties

$$\mathcal{G}_{xy}^2(\omega) \triangleq \frac{|R_{xy}(e^{j\omega})|^2}{R_x(e^{j\omega})R_y(e^{j\omega})}$$

Coherence has many interesting and useful properties

- If $y(n) = h(n) * x(n)$, then $\mathcal{G}_{xy}^2(\omega) = 1$
- $\mathcal{G}_{xy}^2(\omega) = 0$ if
 - $r_{xy}(\ell) = 0$
 - $x(n)$ and $y(n)$ are statistically independent
- Bounded: $0 \leq \mathcal{G}_{xy}^2 \leq 1$
- Symmetry: $\mathcal{G}_{xy}^2(\omega) = \mathcal{G}_{yx}^2(\omega)$
- Can be applied to complex signals, though range must span $-\pi < \omega < \pi$ in that case

Coherence as Correlation

$$\mathcal{G}_{xy}^2(\omega) = \frac{|R_{xy}(e^{j\omega})|^2}{R_x(e^{j\omega})R_y(e^{j\omega})} \quad \rho^2 = \frac{|\text{cov}[x, y]|^2}{\text{var}[x] \text{var}[y]}$$

- Coherence is very similar to the coefficient of determination (square of the correlation coefficient) between two RVs
- It is, actually, the coherence between the random spectra

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j\omega n} dX(e^{j\omega}) \quad \text{E} \left[|dX(e^{j\omega})|^2 \right] = R_x(e^{j\omega}) d\omega$$

$$\mathcal{G}_{xy}^2(\omega) = \frac{|\text{cov} [dX(e^{j\omega}), dY^*(e^{j\omega})]|^2}{\text{var} [dX(e^{j\omega})] \text{var} [dY(e^{j\omega})]}$$

but this requires stochastic integrals, which are beyond the scope of this class (see [?, chapter 4])

- It is therefore worthwhile to understand the difference between correlation of real- and complex-valued RVs

Complex Correlation

Let x and y be complex-valued zero-mean random variables. The complex correlation coefficient is then defined as

$$\rho = \frac{\text{cov}[x, y]}{\sqrt{\text{var}[x] \text{var}[y]}} = \frac{\text{E}[xy^*]}{\sqrt{\text{E}[xx^*] \text{E}[yy^*]}}$$

- ρ will be complex-valued
 - Less intuitive than for real-valued variables
 - Does not retain the sign to indicate positive or negative correlation
- Can't look at scatter plot (x, y span four dimensions)

Understanding Complex Correlation

$$\rho = \frac{E[xy^*]}{\sqrt{E[xx^*] E[yy^*]}}$$

$$E[xy^*] = E[a_x e^{j\theta_x} a_y e^{-j\theta_y}] = E[a_x a_y e^{j(\theta_x - \theta_y)}]$$

$$E[xx^*] = E[a_x^2]$$

- ρ will be zero if $(\theta_x - \theta_y) \sim \mathcal{U}[0, 2\pi]$
- If a_x and a_y are constants, ρ is the **phase correlation**:
 $\rho = E[e^{j(\theta_x - \theta_y)}]$
- If $(\theta_x - \theta_y)$ is a constant, $|\rho|$ is the un-normalized correlation coefficient of the amplitudes

$$\rho = \frac{E[a_x a_y^*]}{\sqrt{E[a_x^2] E[a_y^2]}} e^{j(\theta_x - \theta_y)}$$

- In general is a mixture of phase and amplitude correlation

Understanding Complex Correlation

$$\rho = \frac{E[xy^*]}{\sqrt{E[xx^*] E[yy^*]}}$$

$$E[xy^*] = E[a_x e^{j\theta_x} a_y e^{-j\theta_y}] = E[a_x a_y e^{j(\theta_x - \theta_y)}]$$

$$E[xx^*] = E[a_x^2]$$

- Strong correlation, say $\rho = 0.90$, does not mean could *accurately* estimate x from y or vice versa
 - Sufficient accuracy depends on the application
 - Also does not mean a linear estimate is best
 - Could possibly estimate more accurately with nonlinear estimate
- Weak correlation, say $\rho = 0.10$, does not mean x and y are unrelated
- Only measures degree of *linear* association of x and y

Understanding Complex Correlation Continued

$$R_{\tilde{y}}(e^{j\omega}) = R_y(e^{j\omega}) (1 - \mathcal{G}_{xy}^2(\omega))$$

$$R_{\hat{y}}(e^{j\omega}) = R_y(e^{j\omega}) \mathcal{G}_{xy}^2(\omega)$$

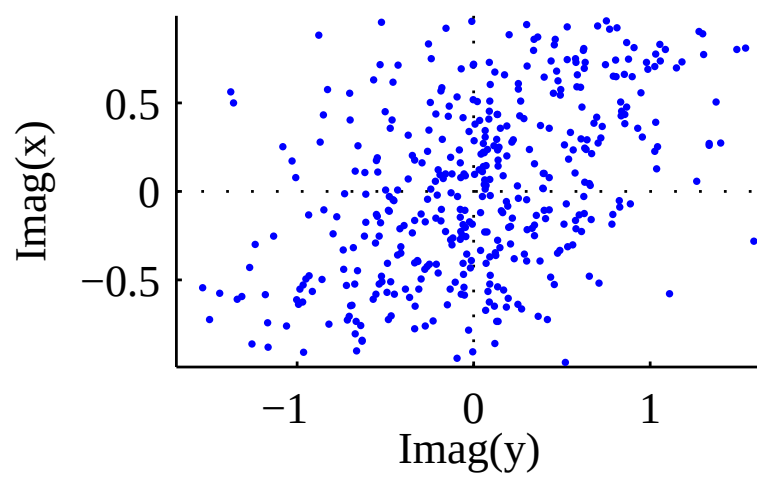
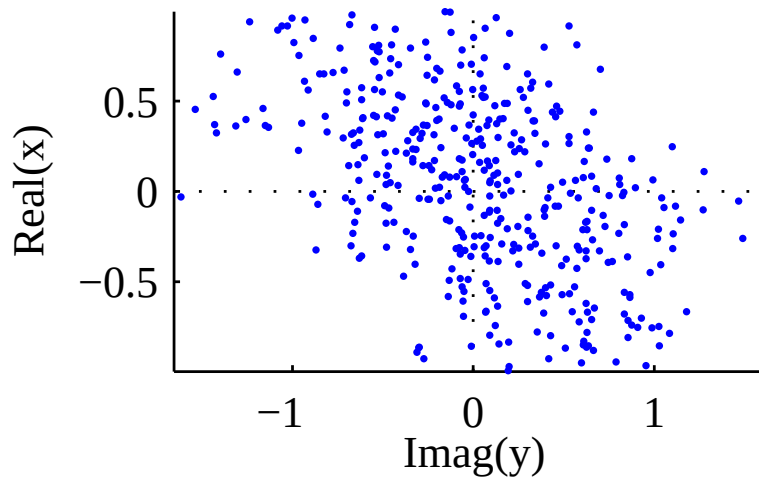
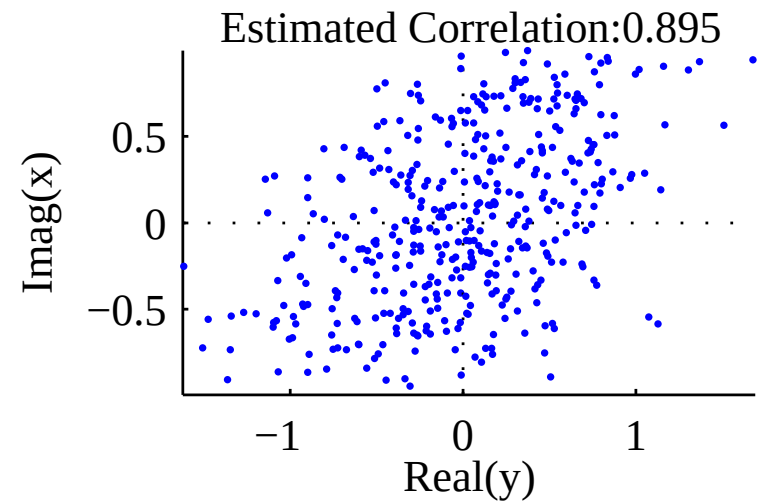
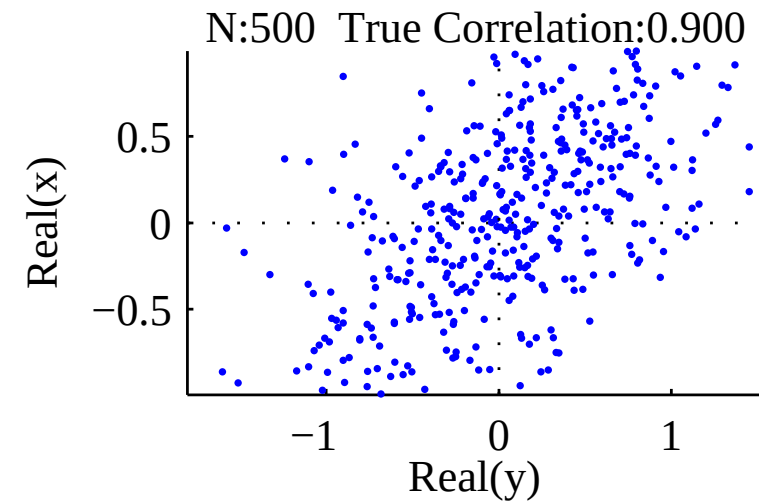
$$R_y(e^{j\omega}) = R_{\hat{y}}(e^{j\omega}) + R_{\tilde{y}}(\omega)$$

- Let $x(n)$ and $y(n)$ be zero-mean jointly WSS random processes
 - $\hat{y}(n) = h(n) * x(n)$ be the best linear estimate of $y(n)$ given $x(n)$
 - $\tilde{y}(n) = y(n) - \hat{y}(n)$
- Then $\mathcal{G}_{xy}^2(\omega)$ can be interpreted as the fraction of the variance explained by an optimal linear model
- Alternatively the MSC is the fraction of $R_y(e^{j\omega})$ that can be estimated from $x(n)$

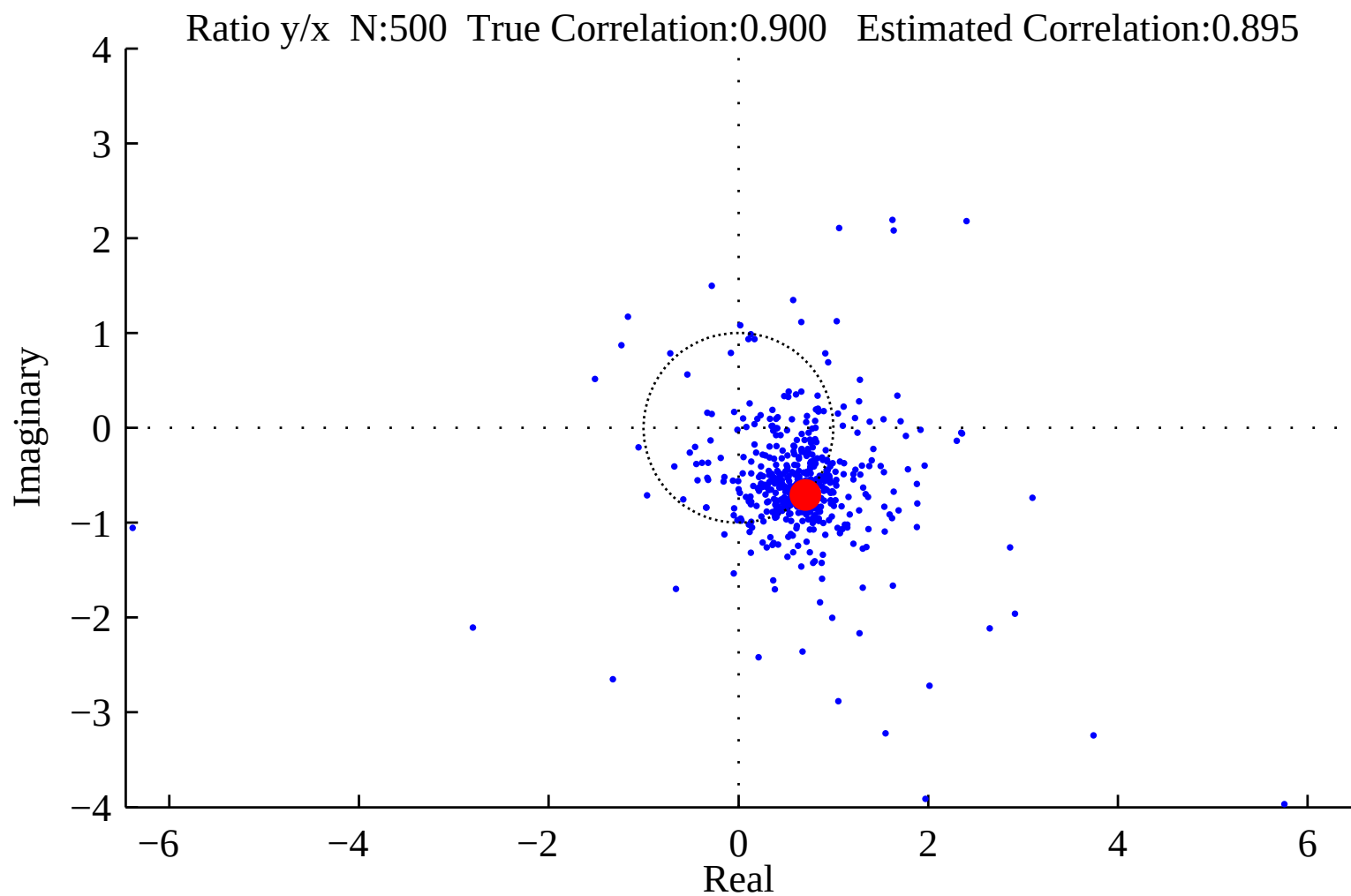
Example 3: Complex Correlation Scatter Plots

Create 500 pairs of IID, zero-mean, unit-variance complex-valued Gaussian RVs, x and y , with a true correlation of $\rho = 0.9$ and $\rho = 0.1$. Estimate the absolute value of the correlation coefficient. Plot scatter plots of the real and imaginary components of x and y . Also create a scatter plot of the ratio of x/y . Discuss the results.

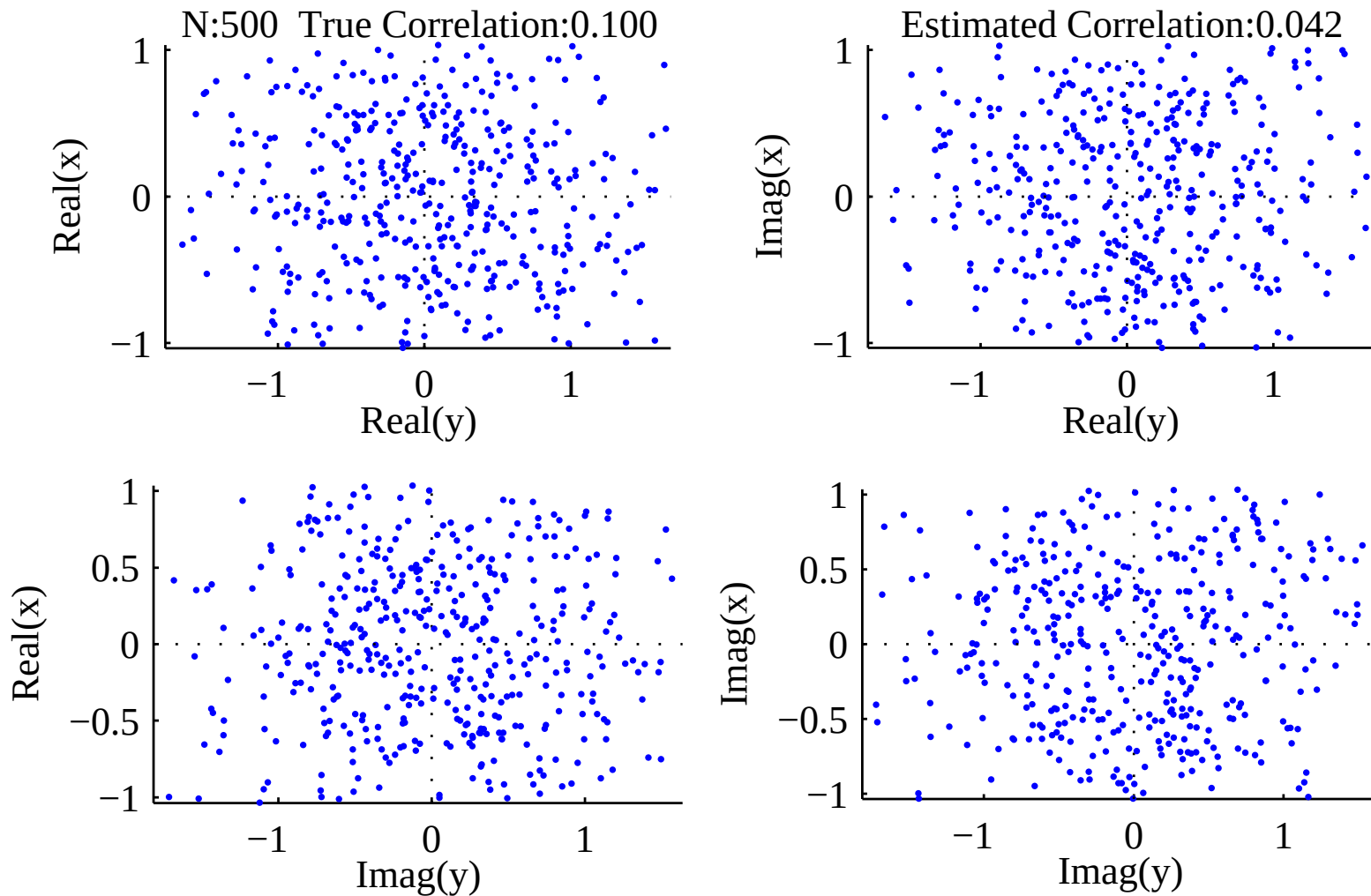
Example 3: Complex Correlation Scatter Plots



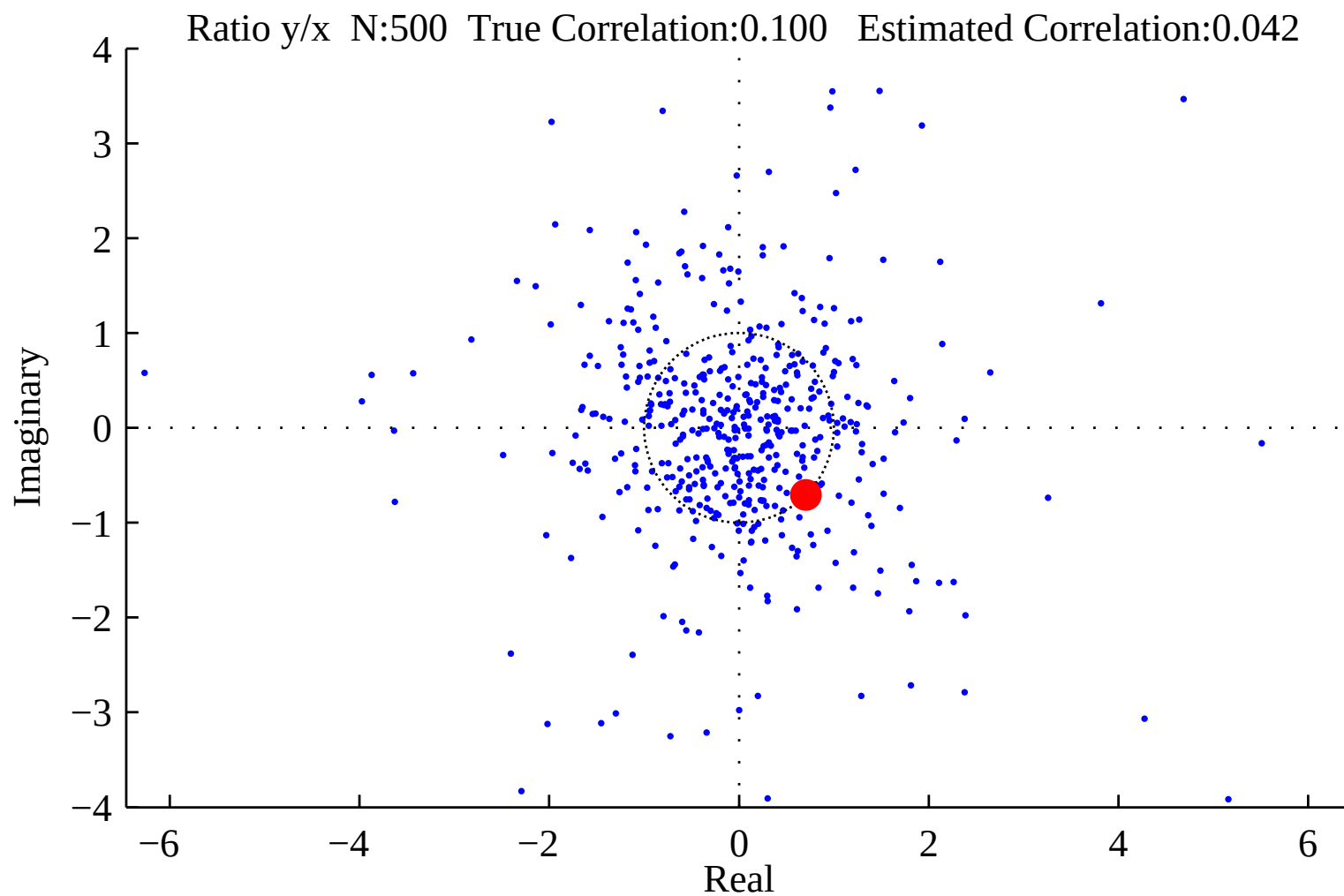
Example 3: Complex Correlation Ratio Scatter Plot



Example 3: Complex Correlation Scatter Plots



Example 3: Complex Correlation Ratio Scatter Plot



Example 3: MATLAB Code

```
%function [] = ComplexCorrelation();
close all;
clear;

N = 500;
R = [0.10 0.90]; % Desired correlation
for c1=1:length(R),
    rho = R(c1);
    am = sqrt(rho^2/(1-rho^2));
    a = am*exp(-j*pi/4);
    x = 1/sqrt(2)*(randn(N,1) + j*randn(N,1));
    w = 1/sqrt(2)*(randn(N,1) + j*randn(N,1));
    y = (w + a*x)./sqrt(1+abs(a)^2);

    yc = conj(y);
    xc = conj(x);

    cyx = sum(y.*xc);
    cyx2 = cyx * conj(cyx);
    sxx2 = sum(x.*xc);
    syy2 = sum(y.*yc);
    Cyx = sqrt(cyx2/(sxx2*syy2)); % Correlation coefficient

    figure;
    FigureSet(1,'LTX');
    subplot(2,2,1);
    h = plot(real(y),real(x),'.');
```

```

        set(h,'MarkerSize',4);
        axis([-1 1 -1 1]*std(y));
        axis('equal');
        AxisLines;
        box off;
        xlabel('Real(y)');
        ylabel('Real(x)');
        title(sprintf('N:%d   True Correlation:%5.3f',N,(abs(a).^2/(1+abs(a)^2)).^(1/2)));
subplot(2,2,2);
    h = plot(real(y),imag(x),'.');
    set(h,'MarkerSize',4);
    axis([-1 1 -1 1]*std(y));
    axis('equal');
    AxisLines;
    box off;
    xlabel('Real(y)');
    ylabel('Imag(x)');
    title(sprintf('Estimated Correlation:%5.3f',Cyx));
subplot(2,2,3);
    h = plot(imag(y),real(x),'.');
    set(h,'MarkerSize',4);
    axis([-1 1 -1 1]*std(y));
    axis('equal');
    AxisLines;
    box off;
    xlabel('Imag(y)');
    ylabel('Real(x)');
subplot(2,2,4);
    h = plot(imag(y),imag(x),'.');
    set(h,'MarkerSize',4);
    axis([-1 1 -1 1]*std(y));

```

```

        axis('equal');
        AxisLines;
        box off;
        xlabel('Imag(y)');
        ylabel('Imag(x)');
AxisSet(8);
%print -dtiff -r150 ../ComplexCorrelationScatterPlots;
print(sprintf('ComplexCorrelationScatterPlots%03d',rho*100),'-depsc');

figure;
FigureSet(2,'LTX');
Circle;
hold on;
h = plot(real(y./x),imag(y./x),'.',real(a)./abs(a),imag(a)./abs(a),'ro');
set(h(1),'MarkerSize',4);
set(h(2),'MarkerFaceColor','r');
hold off;
axis([-2 2 -2 2]*2.0);
axis('equal');
xlim(xlim-mean(xlim)); % Center x-axis
AxisLines;
box off;
xlabel('Real');
ylabel('Imaginary');
title(sprintf('Ratio y/x  N:%d  True Correlation:%5.3f  Estimated Correlation:%5.3f',N,rho,Cyx));
AxisSet(8);
print(sprintf('ComplexCorrelationRatioScatterPlots%03d',rho*100),'-depsc');
end;

```

Estimation

Let $x(n)$ be a WSS process

- It seems that most people use Welch's nonparametric PSD and CPSD estimates (why?)
- Most of the statistical properties are apparently only known for Bartlett's estimate
- Usual (unrealistic) assumptions apply
 - Independent segments
 - Gaussian process
 - No spectral leakage

Signal Plus Noise Example

Let $x(n)$ and $w(n)$ be zero-mean WGN processes with variance σ^2 .

$$y(n) = x(n) + \frac{1}{a}w(n)$$

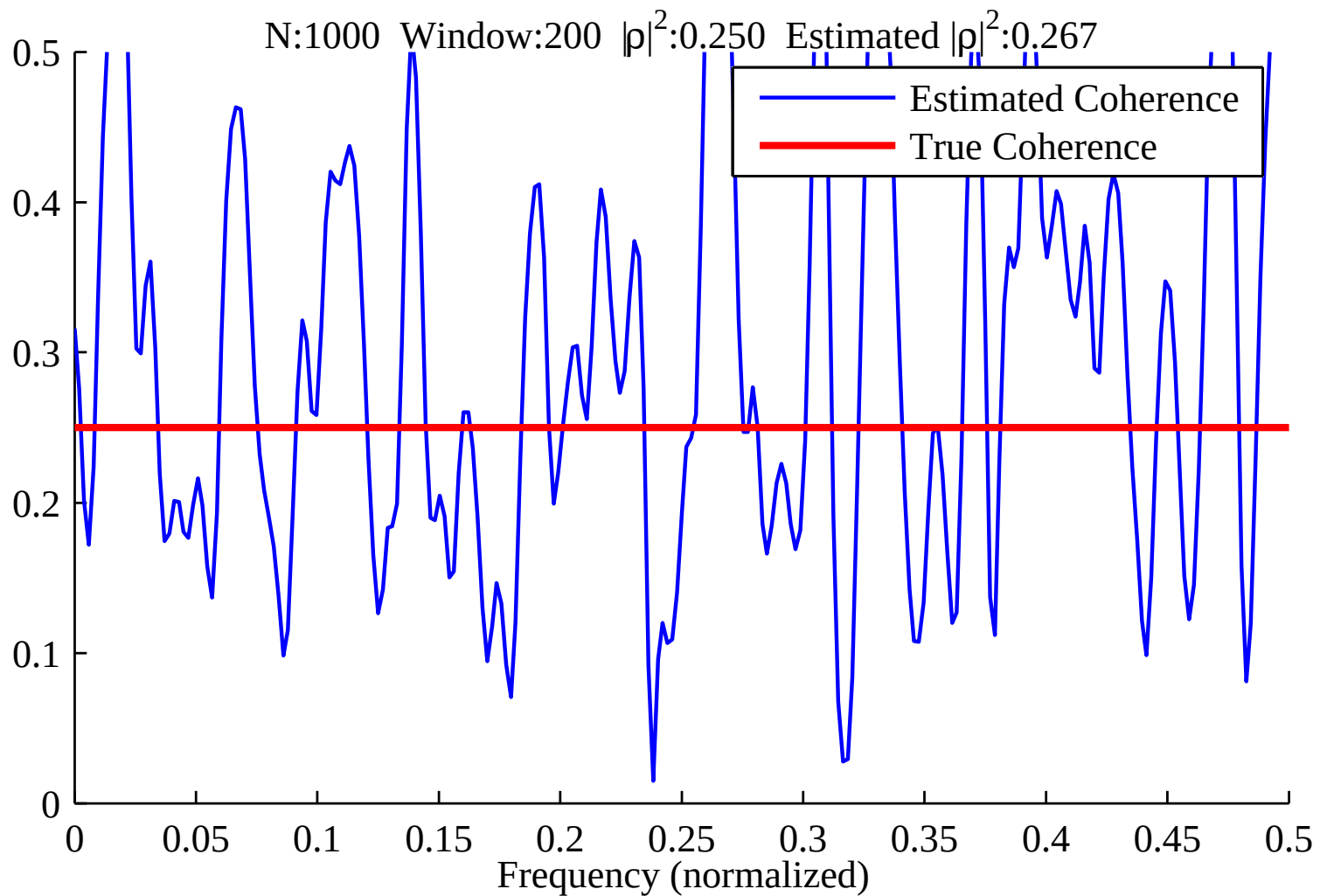
$$\begin{aligned}\mathcal{G}^2(\omega) &= \frac{R_x(e^{j\omega})}{R_x(e^{j\omega}) + \frac{1}{a^2}R_w(e^{j\omega})} \\ &= \frac{a^2\sigma_x^2}{a^2\sigma_x^2 + \sigma_w^2} \\ &= \frac{a^2}{a^2 + 1}\end{aligned}$$

- Since all signals are WGN, the PSDs are constant and spectral leakage does not cause any bias

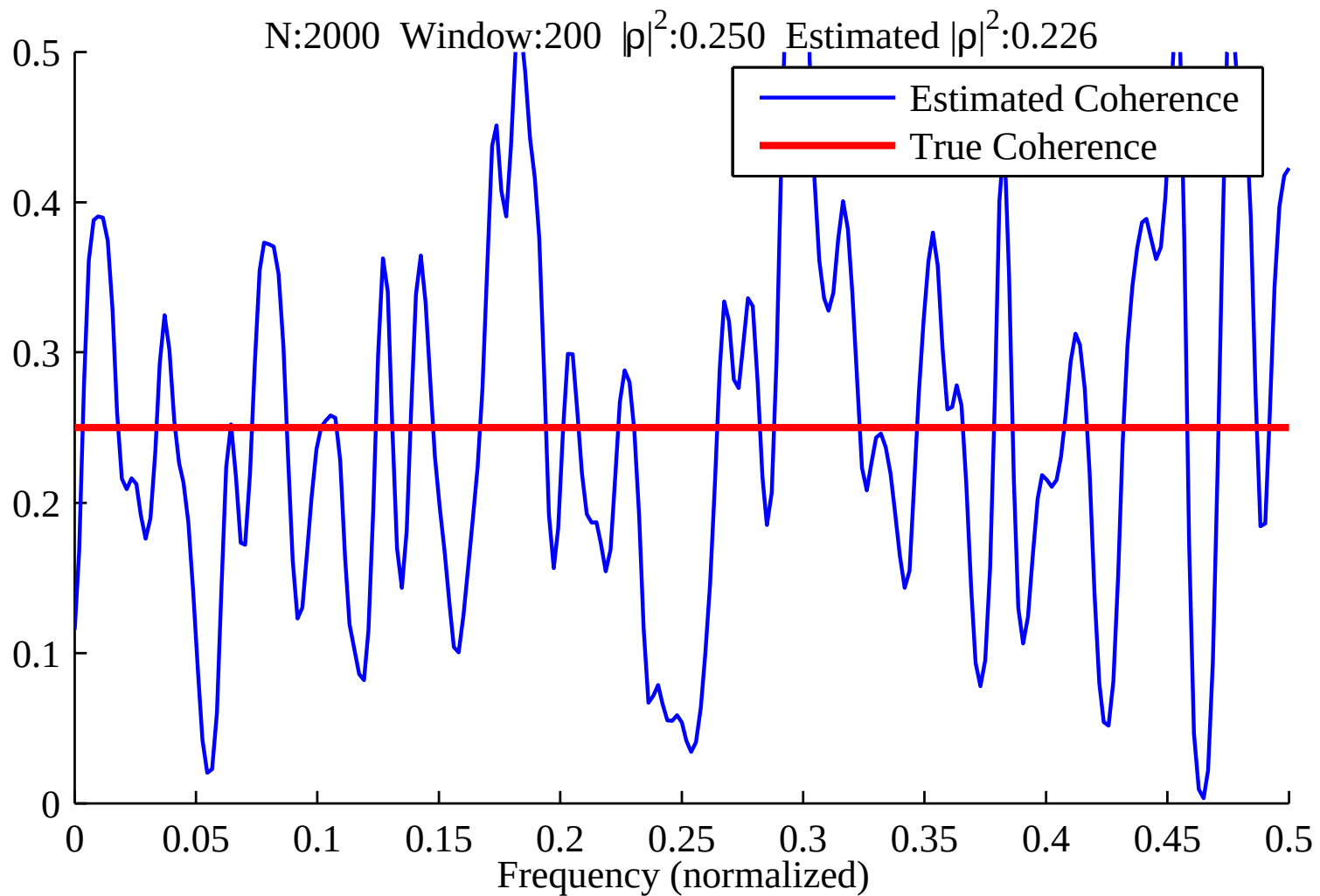
Example 4: White Noise Coherence

Create a single realization of $N = 1000, 2000, 5000$, and $10,000$ samples of the random process $y(n) = x(n) + \frac{1}{a}w(n)$ where $x(n)$ and $w(n)$ be zero-mean WGN processes with variance σ^2 . Plot the true and estimated coherence for a window length of 200 samples. What impact does increasing N have on K . What is the effect on $\hat{\mathcal{G}}_{xy}^2(\omega)$?

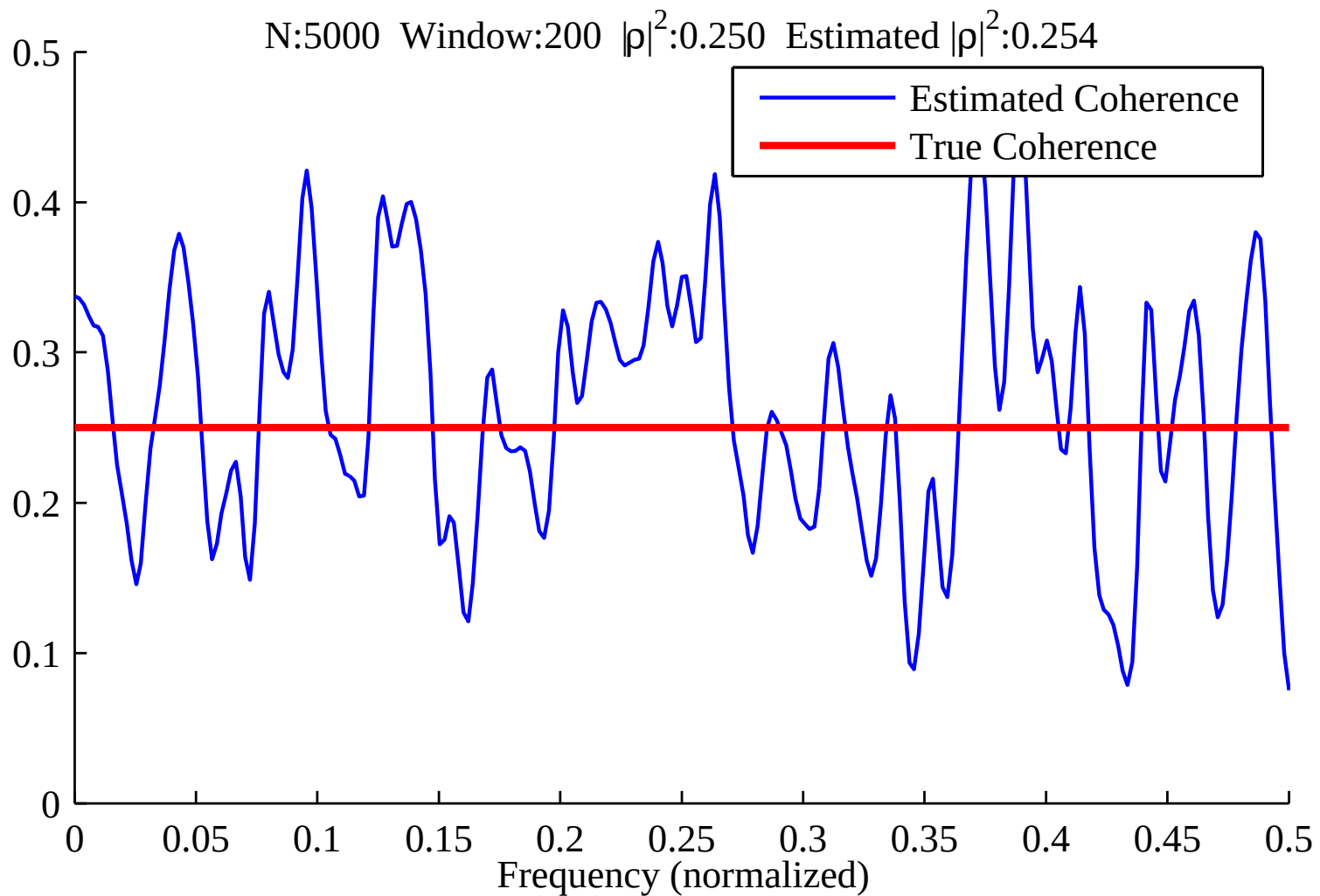
Example 4: White Noise Coherence



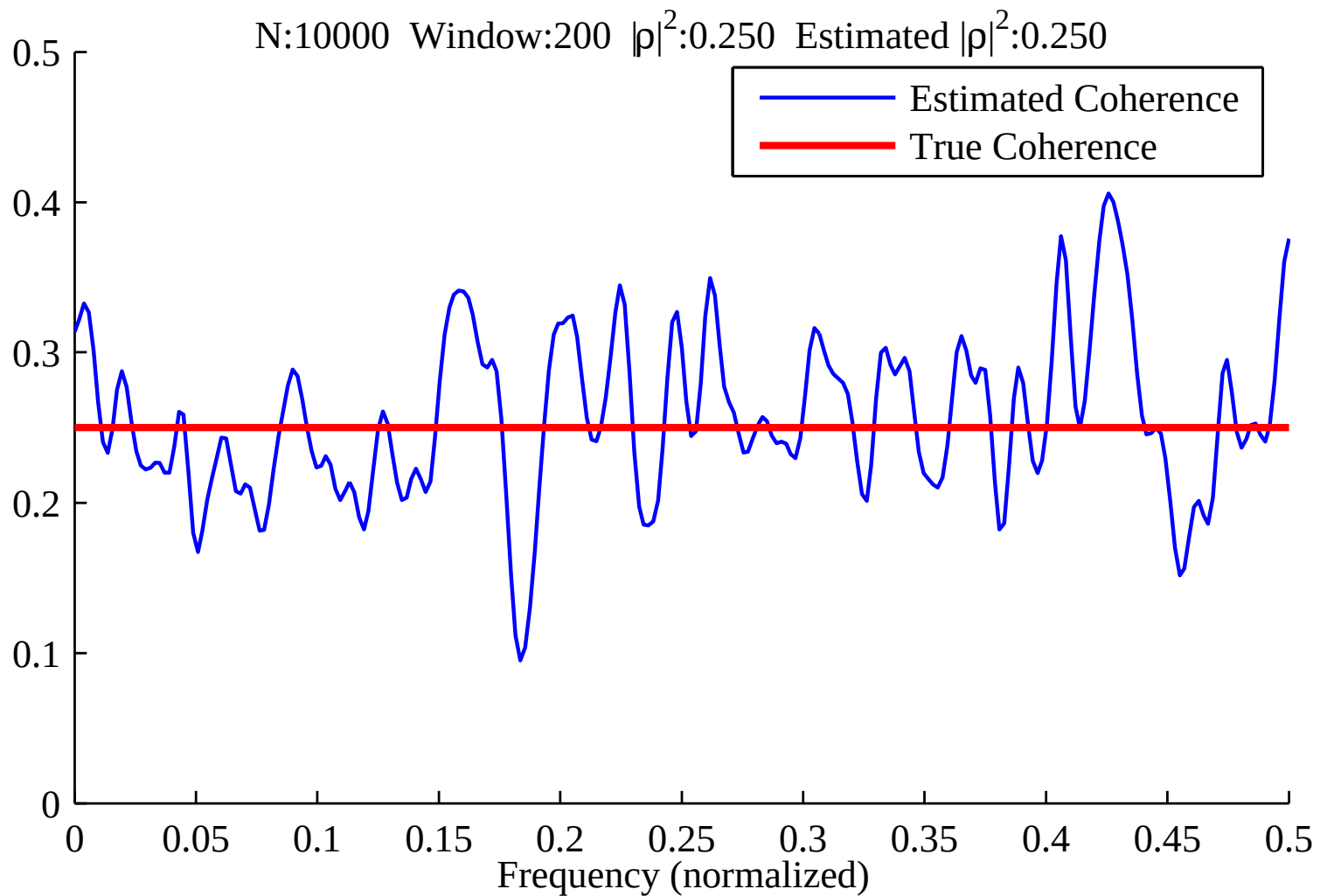
Example 4: White Noise Coherence



Example 4: White Noise Coherence



Example 4: White Noise Coherence



Example 4: MATLAB Code

```
%function [] = SignalPlusNoise;
close all;
clear;

N    = [1000,2000,5000,10000];
wl   = 200;
rho  = 0.5; % Desired correlation
a    = sqrt(rho^2/(1-rho^2));

for c1=1:length(N),
    x    = randn(N(c1),1);
    w    = randn(N(c1),1);
    y    = (w + a*x);

    r    = sum(x.*y)./sqrt(sum(x.^2)*sum(y.^2));

    [c,f] = Coherency(x,y,1,wl);
    FigureSet(1,'LTX');
    h = plot(f,c.^2);
    set(h,'LineWidth',0.8);
    hold on;
        h = plot([min(f) max(f)],rho^2*[1 1],'r');
        set(h,'LineWidth',1.5);
    hold off;
    xlabel('Frequency (normalized)');
    title(sprintf('N:%d Window:%d |\\rho|^2:%5.3f Estimated |\\rho|^2:%5.3f',N(c1),wl,rho^2,r^2));
    xlim([0 0.5]);
```

```
ylim([0 2*rho.^2]);  
box off;  
AxisSet(8);  
legend('Estimated Coherence','True Coherence');  
print(sprintf('SignalPlusNoise%02d',N(c1)/1000),'-depsc');  
end;
```

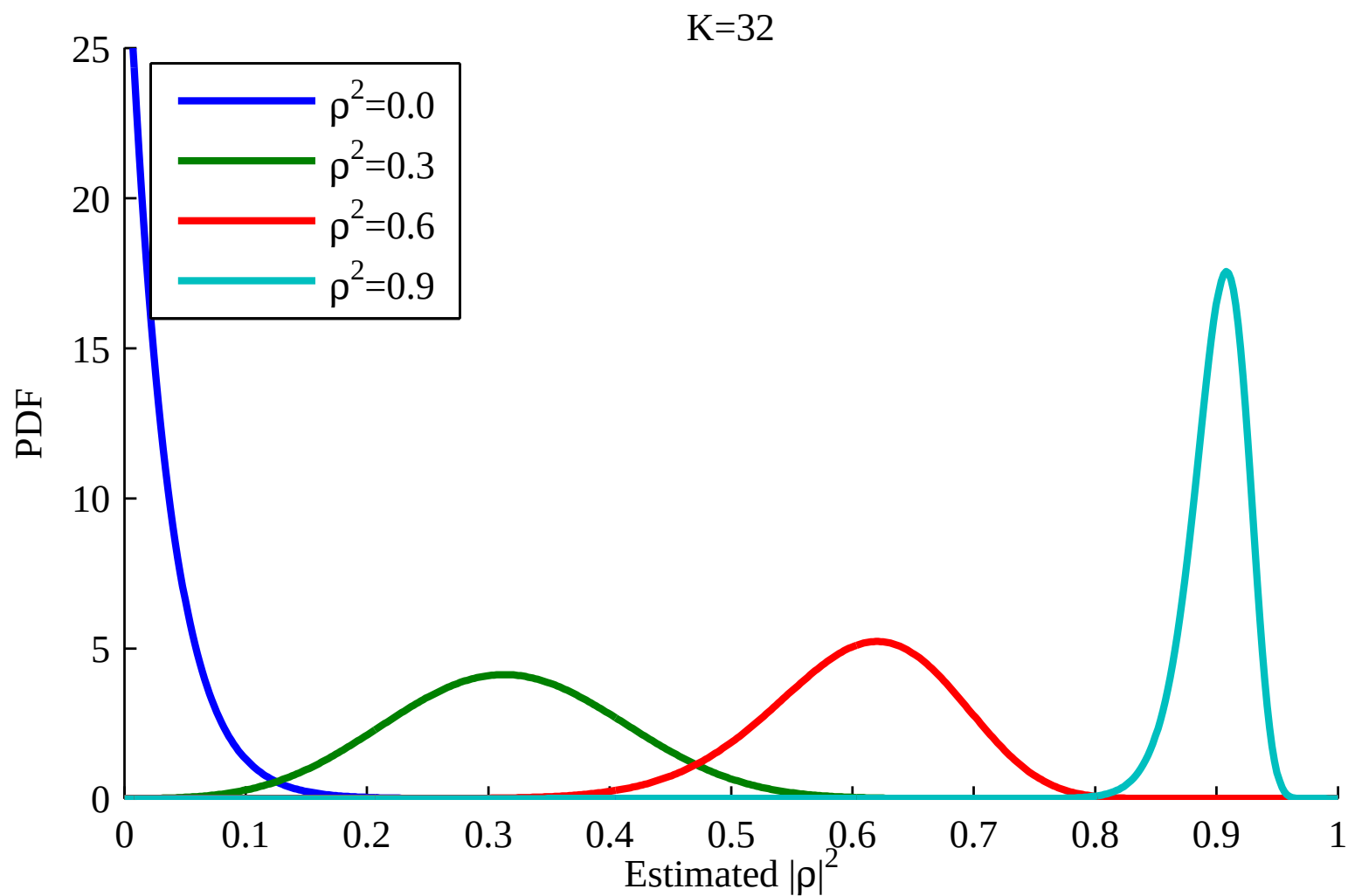
Sampling Distribution

$$f(|\hat{\rho}|^2; K, |\rho|^2) = (K-1)(1-|\rho|^2)^K (1-|\hat{\rho}|^2)^{K-2} {}_2F_1(K, K; 1; |\rho|^2|\hat{\rho}|^2)$$

for $0 \leq |\rho|^2|\hat{\rho}|^2 < 1$

- When x and y are K samples of an IID, zero-mean Gaussian RVs, the distribution of $|\hat{\rho}|^2$ is known
- This same PDF is usually used for $\mathcal{G}_{xy}^2(\omega)$
- Resulting PDF is only approximate for MSC because these assumptions don't hold for $\mathcal{G}_{xy}^2(\omega)$
 - Segments are not independent, even with Bartlett's method
 - Spectral leakage
- ${}_2F_1(K, K; 1; |\rho|^2|\hat{\rho}|^2)$ is a hypergeometric function (see hypergeom in MATLAB)

Example 5: Coherence PDF



Example 5: MATLAB Code

```
clear;
close all;

K = 32; % Disjoint degrees of freedom
c = [0.0 0.3 0.6 0.9]; % True coherence values
nc = length(c);
x = linspace(0,1,500);
F = zeros(nc,length(x));
for c1=1:nc,
    F(c1,:) = (K-1)*(1-c(c1))^K*(1-x).^(K-2).*hypergeom([K K],1,c(c1)*x);
    lb{c1} = sprintf('\rho^2=%3.1f',c(c1));
end;

figure;
FigureSet(1,'LTX');
h = plot(x,F);
set(h,'LineWidth',1.5);
box off;
xlabel('Estimated |\rho|^2');
ylabel('PDF');
title(sprintf('K=%d',K));
xlim([0 1]);
ylim([0 25]);
AxisSet(8);
legend(lb,'Location','NorthWest');
print('CoherencePDF','-depsc');
```

Coherence Observations

$$f(|\hat{\rho}|^2; K, |\rho|^2) = (K-1)(1-|\rho|^2)^K (1-|\hat{\rho}|^2)^{K-2} {}_2F_1(K, K; 1; |\rho|^2|\hat{\rho}|^2)$$

for $0 \leq |\rho|^2|\hat{\rho}|^2 < 1$

- PDF is skewed
 - Right for $|\rho|^2$ near 0
 - Left for $|\rho|^2$ near 1
- Widest (most variable) near $|\rho|^2 = 0.3$
- Narrows (less variance) as K increases (not shown)

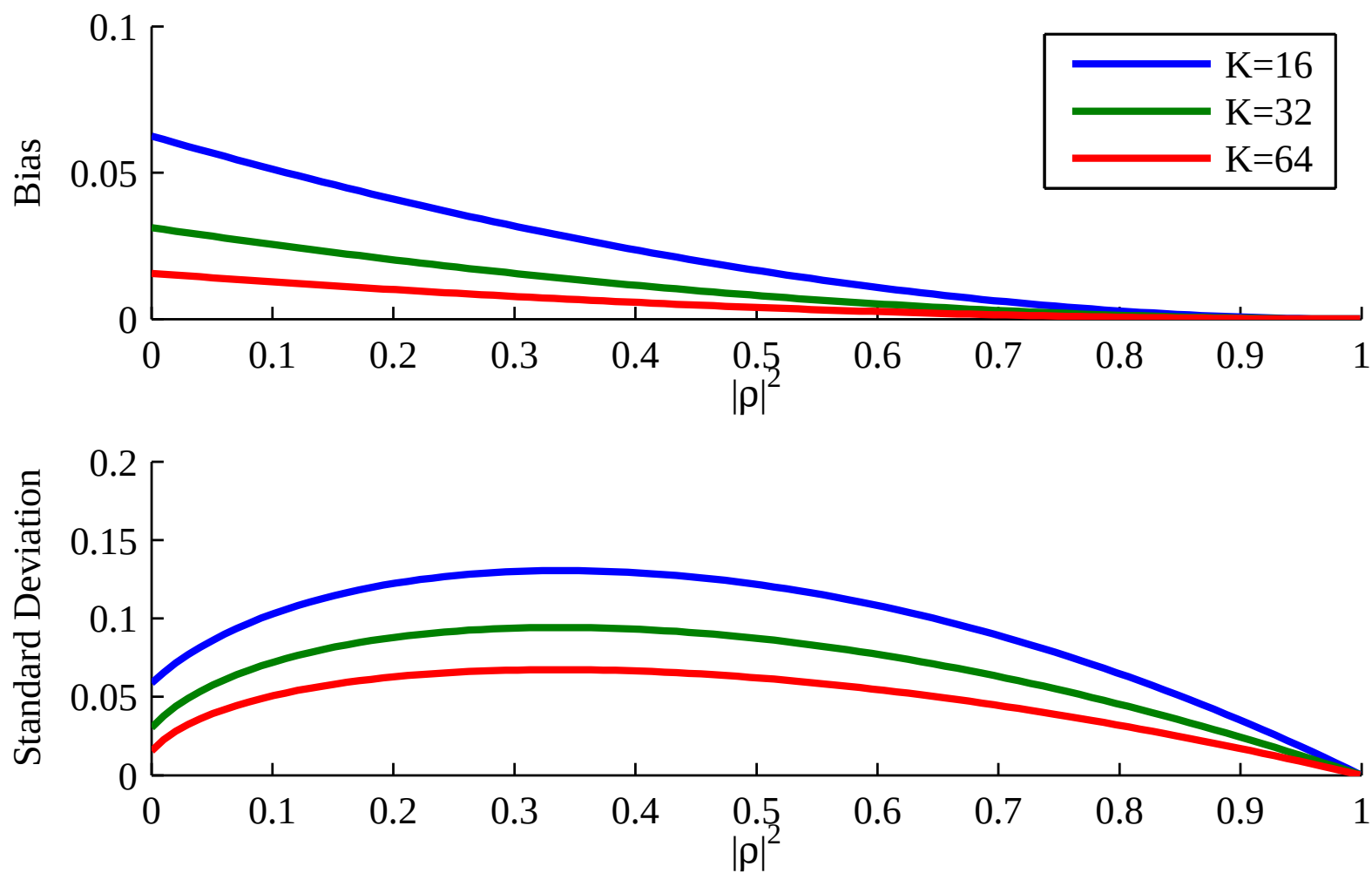
Bias and Variance

$$\begin{aligned}\text{bias}[\rho, \hat{\rho}] &= \text{E}[|\hat{\rho}|^2 | K, |\rho|^2] - |\rho|^2 \\ &= \frac{(1 - |\rho|^2)^K}{K} {}_3F_2(2, K, K; K + 1, 1; |\rho|^2) - |\rho|^2\end{aligned}$$

$$\begin{aligned}\text{var} [|\hat{\rho}|^2] &= \text{E} [|\hat{\rho}|^4] - \text{E} [|\hat{\rho}|^2]^2 \\ &= \frac{2(1 - |\rho|^2)^K}{K(K + 1)} {}_3F_2(3, K, K; K + 2, 1; |\rho|^2) \\ &\quad - \left[\frac{(1 - |\rho|^2)^K}{K} {}_3F_2(2, K, K; K + 1, 1; |\rho|^2) \right]^2\end{aligned}$$

- The bias and variance are known also for independent Gaussian segments
- Bias does not include the effects of spectral leakage or time delay

Example 6: Coherence Bias and Variance



Example 6: MATLAB Code

```
clear;
close all;

K = [16 32 64];          % No. independent segments
nK = length(K);
c = linspace(0,0.999,100);
nc = length(c);
B = zeros(nK,nc);
V = zeros(nK,nc);
for c1=1:nK,
    k = K(c1);
    B(c1,:) = hypergeom([2 k k],[k+1 1],c).*(1-c).^k/k - c;
    V(c1,:) = hypergeom([3 k k],[k+2 1],c).*2.*(1-c).^k/(k*(k+1)) - (hypergeom([2 k k],[k+1 1],c).*(1-c).^k/k - c);
    lb{c1} = sprintf('K=%d',k);
end;

figure;
FigureSet(1,'LTX');
subplot(2,1,1);
h = plot(c,B);
set(h,'LineWidth',1.5);
box off;
xlabel('| \rho |^2');
ylabel('Bias');
xlim([0 1]);
AxisSet(8);
legend(lb,'Location','NorthEast');
```

```
subplot(2,1,2);  
    h = plot(c,V.^(0.5));  
    set(h,'LineWidth',1.5);  
    box off;  
    xlabel('\rho|^2');  
    ylabel('Standard Deviation');  
    xlim([0 1]);  
    AxisSet(8);  
print('CoherenceBiasVariance','-depsc');
```

Bias and Variance Properties

- Bias is largest when $|\rho|^2 = 0$ and smallest when $|\rho|^2 = 1$
- Variance largest when $|\rho|^2 = 1/3$ and smallest when $|\rho|^2 = 1$
- When $|\rho|^2 = 1$, the estimate is perfect! (no bias or variance)
- Both decrease with K
- The estimate is asymptotically unbiased and is consistent

Bias Variance Tradeoff

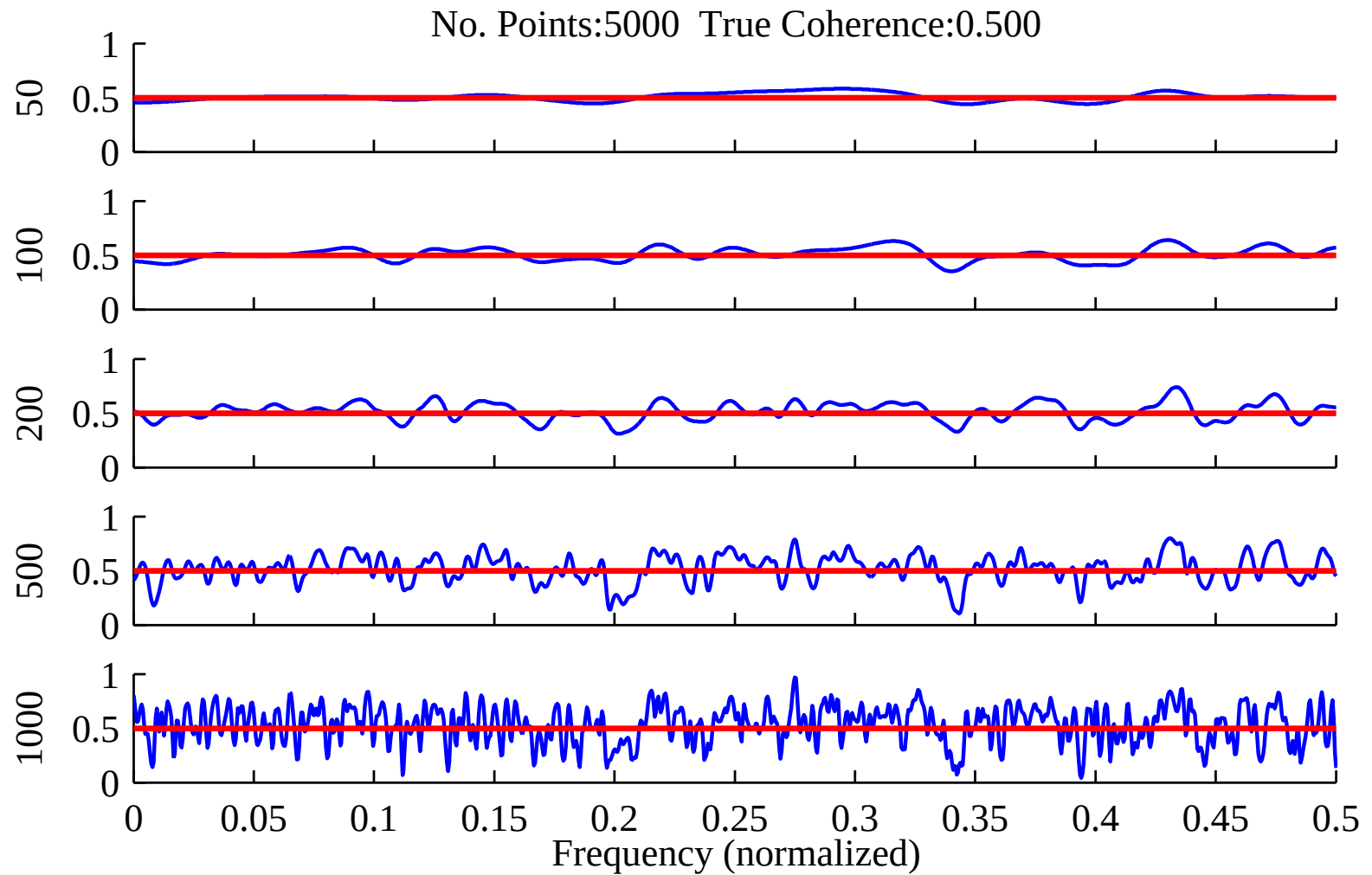
- As usual, there is a bias-variance tradeoff
- With Bartlett and Welch's methods increasing K decreases bias and variance, but increases spectral leakage
- For a fixed overlap, small K
 - Better spectral resolution
 - Low bias
 - Large variance
- For a fixed overlap, large K
 - Worse resolution
 - More biased (spectral leakage, smoothing)
 - Less variability
- Same tradeoff applies to L for the Blackman-Tukey method
- Can decrease bias and variance with Welch's method
- Improvement saturates with 62.5% overlap

Example 7: Effect of K

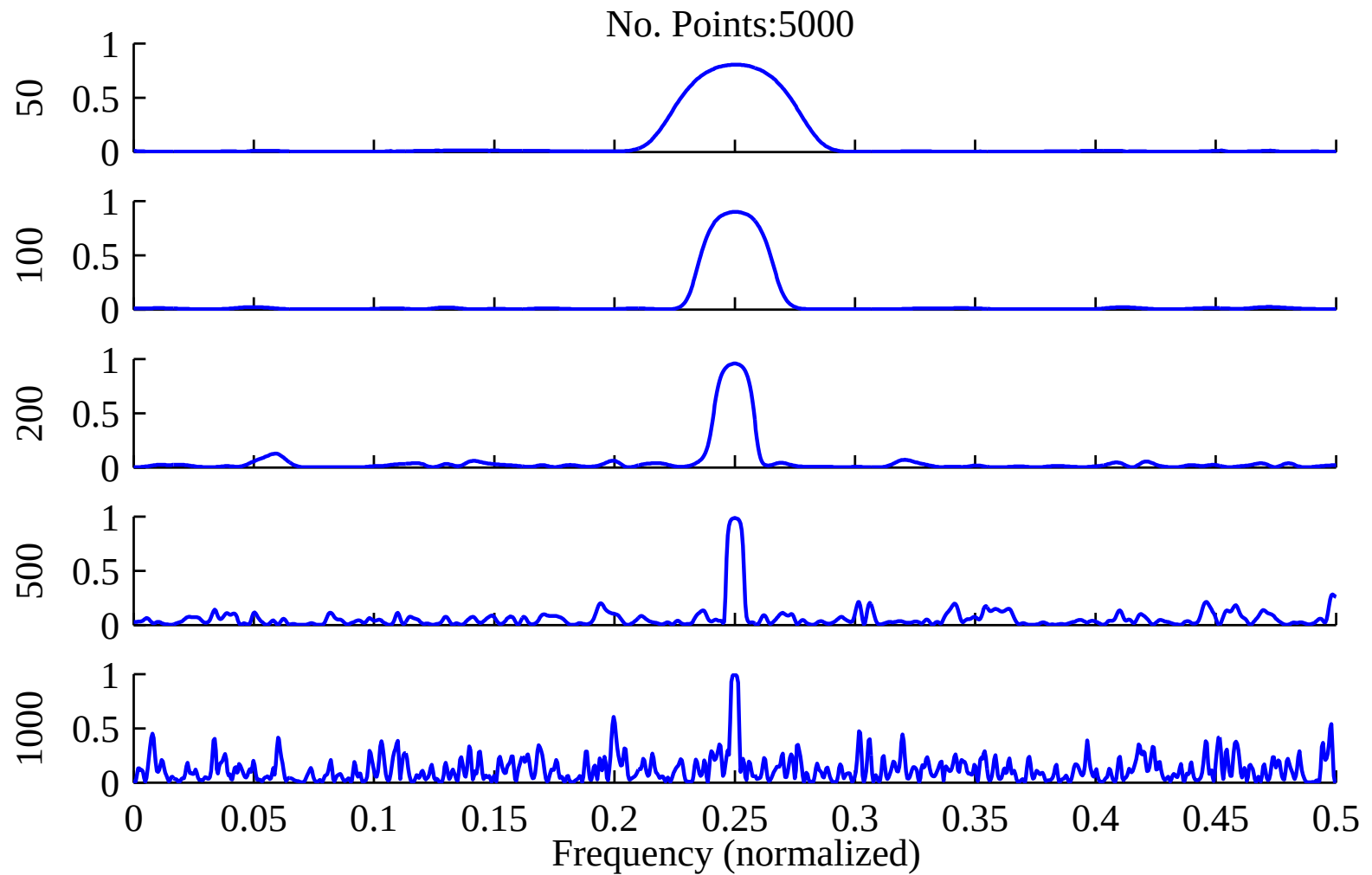
Create a single realization of $N = 5000$ samples of the random process $y(n) = x(n) + \frac{1}{a}w(n)$ where $x(n)$ and $w(n)$ be zero-mean WGN processes with variance σ^2 . Plot the true and estimated coherence for a window lengths of 100, 500, 200, 100, and 50 samples. Use a fixed overlap of 50%. Repeat for $x(n) = \cos(\pi/2n)$.

- How does K scale with the window length?
- What is the effect of increasing K ?

Example 7: Coherence Bias and Variance



Example 7: Coherence Bias and Variance



Example 7: MATLAB Code

```
function [] = WindowLengths;
close all;

np = 5000;
nf = 2^10;
wl = 200;
rho = sqrt(0.50); % Desired correlation
a = sqrt(rho^2/(1-rho^2));
w = randn(np,1);

wl = [50 100 200 500 1000];
nw = length(wl);

%=====
% Sinusoid Plus Noise
%=====
n = (1:np)';
x = sqrt(0.5)*cos(2*pi*0.25*n);
y = (w + a*x);

figure;
FigureSet(1,'LTX');
for cnt = 1:nw,
    subplot(nw,1,cnt);
    [c,f] = Coherency(x,y,1,wl(cnt),[],50,nf);
    h = plot(f,c.^2);
    set(h(1),'LineWidth',0.8);
```

```

ylim([0 1]);
xlim([0 0.5]);
box off;
ylabel(sprintf('%d',wl(cnt)));
if cnt<nw,
    set(gca,'XTickLabel','');
else
    xlabel('Frequency (normalized)');
end;
if cnt==1,
    title(sprintf('No. Points:%d ',np));
end;
end;
AxisSet(8);
print('WindowLengthsSinusoidalNoise','-depsc');

%=====
% Noise Plus Noise
%=====
x = randn(np,1);
y = (w + a*x);

figure;
FigureSet(2,'LTX');
for cnt = 1:nw,
    subplot(nw,1,cnt);
    [c,f] = Coherency(x,y,1,wl(cnt),[],50,nf);
    h = plot(f,c.^2);
    set(h(1),'LineWidth',0.8);
    ylim([0 1]);
    xlim([0 0.5]);

```

```

hold on;
    h = plot(xlim,0.5*[1 1], 'r');
    set(h, 'LineWidth', 1.2);
    hold off;
box off;
ylabel(sprintf('%d', wl(cnt)));
if cnt~=nw,
    set(gca, 'XTickLabel', '');
else
    xlabel('Frequency (normalized)');
end;
if cnt==1,
    title(sprintf('No. Points:%d True Coherence:%5.3f', np, rho^2));
end;
end;
AxisSet(8);
print('WindowLengthsNoiseNoise', '-depsc');

```

Confidence Intervals

$$F(|\hat{\rho}|^2; K, |\rho|^2) = |\hat{\rho}|^2 \left(\frac{1 - |\rho|^2}{1 - |\rho|^2 |\hat{\rho}|^2} \right)^K \times \sum_{k=0}^{K-2} \left(\frac{1 - |\hat{\rho}|^2}{1 - |\rho|^2 |\hat{\rho}|^2} \right)^k {}_2F_1(-k, 1 - K; 1; |\rho|^2 |\hat{\rho}|^2)$$

- Given all the usual assumptions, the PDF and cdf depend only on K and ρ
- K is known (user-specified)
- If ρ was known, could calculate exact confidence intervals
- Classic problem: true PDF depends on the parameter we're trying to estimate
- There's a trick this time, though

Confidence Intervals Continued

Define $a_L(\rho)$ and $a_U(\rho)$ as the $\alpha/2$ and $1 - \alpha/2$ percentiles of $F(|\hat{\rho}|^2; K, |\rho|^2)$, respectively

$$\Pr \{ \hat{\rho} < a_L(\rho) | \rho \} = \alpha/2 = F(a_L; K, |\rho|^2)$$

$$\Pr \{ \hat{\rho} > a_U(\rho) | \rho \} = \alpha/2 = 1 - F(a_U; K, |\rho|^2)$$

Then

$$\Pr \{ a_L(\rho) < \hat{\rho} < a_U(\rho) | \rho \} = 1 - \alpha = F(a_U; K, |\rho|^2) - F(a_L; K, |\rho|^2)$$

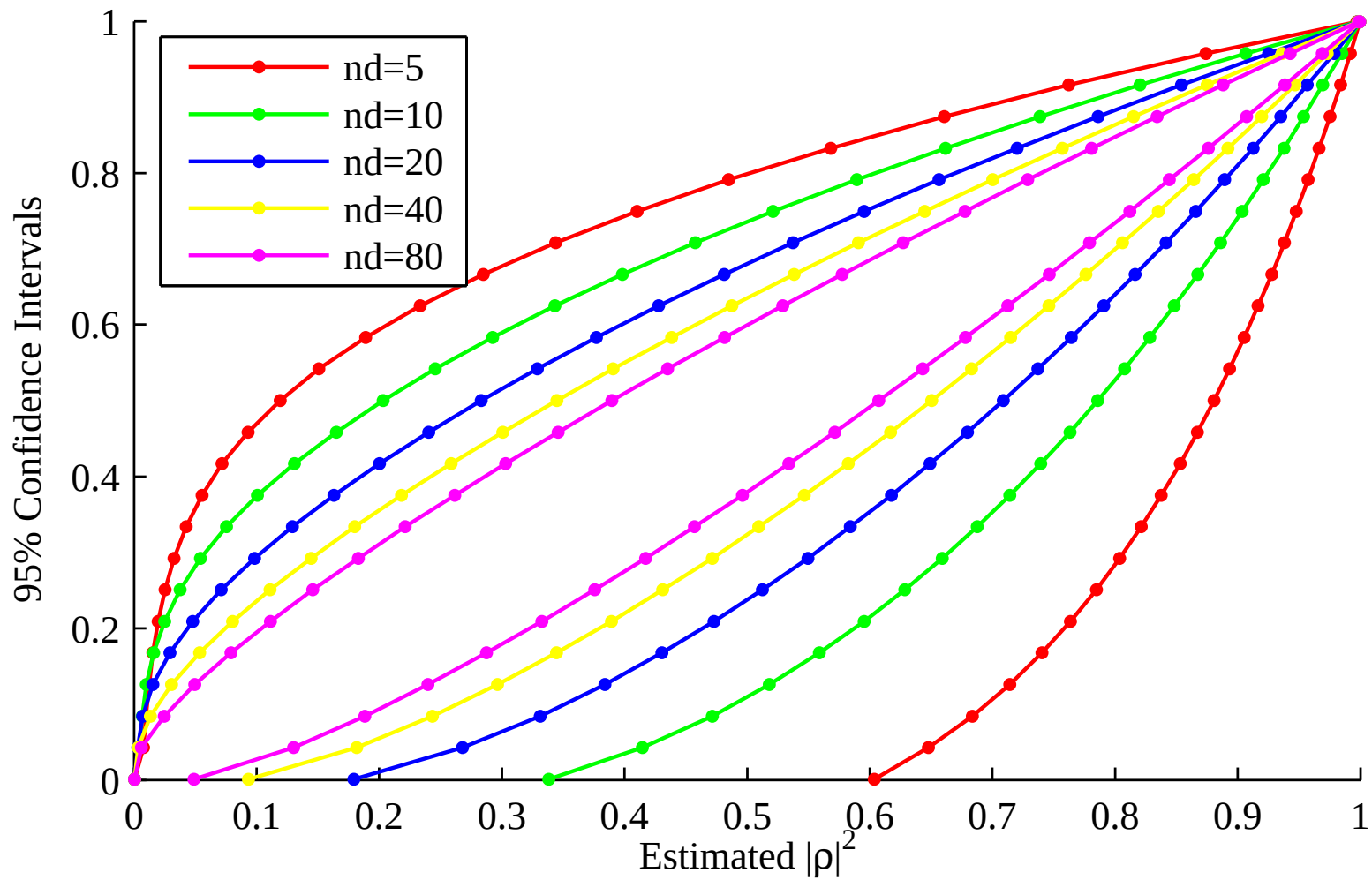
- We do not have a closed form expression for $a_L(\rho)$ or $a_U(\rho)$, but we can solve for them numerically
- They both increase monotonically with ρ , as expected
- Thus, they are both invertible

Confidence Intervals Continued

$$\begin{aligned} 1 - \alpha &= \Pr \{a_L(\rho) < \hat{\rho} < a_U(\rho) | \rho\} \\ &= \Pr \{\rho < a_L^{-1}(\hat{\rho}), a_U^{-1}(\hat{\rho}) < \rho | \rho\} \\ &= \Pr \{a_U^{-1}(\hat{\rho}) < \rho < a_L^{-1}(\hat{\rho}) | \rho\} \end{aligned}$$

- We can create an upper and lower confidence interval by inverting the appropriate percentile functions
- Cannot be done in closed form, must be done numerically
- Only applies under all the usual assumptions
- Since the assumptions are not met in practice, these are merely approximations

Example 8: 95% Confidence Intervals



Example 8: MATLAB Code

```
function [] = CoherenceCI();

nd = [5 10 20 40 80];
nc = 25;

CI = zeros(length(nd),nc,3);
for c0=1:length(nd),
    ci = CoherenceCIs(nd(c0),nc);
    CI(c0,:,:)= ci;
end;

figure;
FigureSet(1,'LTX');
ca = {'r','g','b','y','m','c'};
ha = [];
for c1=1:length(nd),
    ci = squeeze(CI(c1,:,:));
    h = plot(ci(:,1),ci(:,3),ci(:,2),ci(:,3));
    hold on;
    set(h,'Color',ca{c1});
    set(h,'Marker','.');
    set(h,'MarkerSize',8);
    set(h,'LineWidth',0.8);
    ha = [ha;h(1)];
    lg{c1} = sprintf('nd=%d',nd(c1));
end;
hold off;
```

```

box off;
xlabel('Estimated  $|\rho|^2$ ');
ylabel('95% Confidence Intervals');
xlim([0 1]);
ylim([0 1]);
AxisSet(8);
legend(ha,lg,'Location','NorthWest');
print('CoherenceCI','-depsc');

function [CI] = CoherenceCIs(nd,nc,llb,uub);
    tol = 0.001; % Tolerance of solution (resolution)
    los = 0.05; % Level of significance
    mxi = 50; % Maximum number of iterations

    op = optimset('TolX',tol,'Display','Off','MaxIter',mxi);
    msc = linspace(0.001,0.999,nc)';

    CI = zeros(nc,3);

    for c1=1:nc,
        [al,po,ef,ot] = fminbnd(@(x)( los/2-CoherenceCDF(x,nd,msc(c1))).^2,0,msc(c1),op);
        if ef~=1,warning(ot.message);end;
        [au,po,ef,ot] = fminbnd(@(x)(1-los/2-CoherenceCDF(x,nd,msc(c1))).^2,msc(c1),1,op);
        if ef~=1,warning(ot.message);end;
        CI(c1,1) = al;
        CI(c1,2) = au;
        CI(c1,3) = msc(c1);
    end;
end

function [p] = CoherenceCDF(ch,nd,c);

```

```

p = 0;
for k=0:nd-2,
    p = p + ((1-ch)./(1-ch*c)).^k.*hypergeometric21(k,nd,ch*c);
end;
p = p.*ch.*((1-c)./(1-c.*ch)).^nd;
end

function [x] = hypergeometric21(k,nd,p);
t = zeros(k+1,1);
t(1) = 1;
for c2=2:k+1,
    i = c2-1;
    t(c2) = t(c2-1)*((c2-2-k)*(c2-1-nd)*p)/((c2-1)^2);
end;
x = sum(t);
end
end

```

Sources of Bias

- The estimate is intrinsically biased
 - Estimate is bounded $0 \leq |\hat{\rho}|^2 \leq 1$
 - If $|\rho|^2 = 1$ or $|\rho|^2 = 0$ then variability causes bias
- Bias caused by delay
- Bias caused by spectral leakage

Spectral Leakage

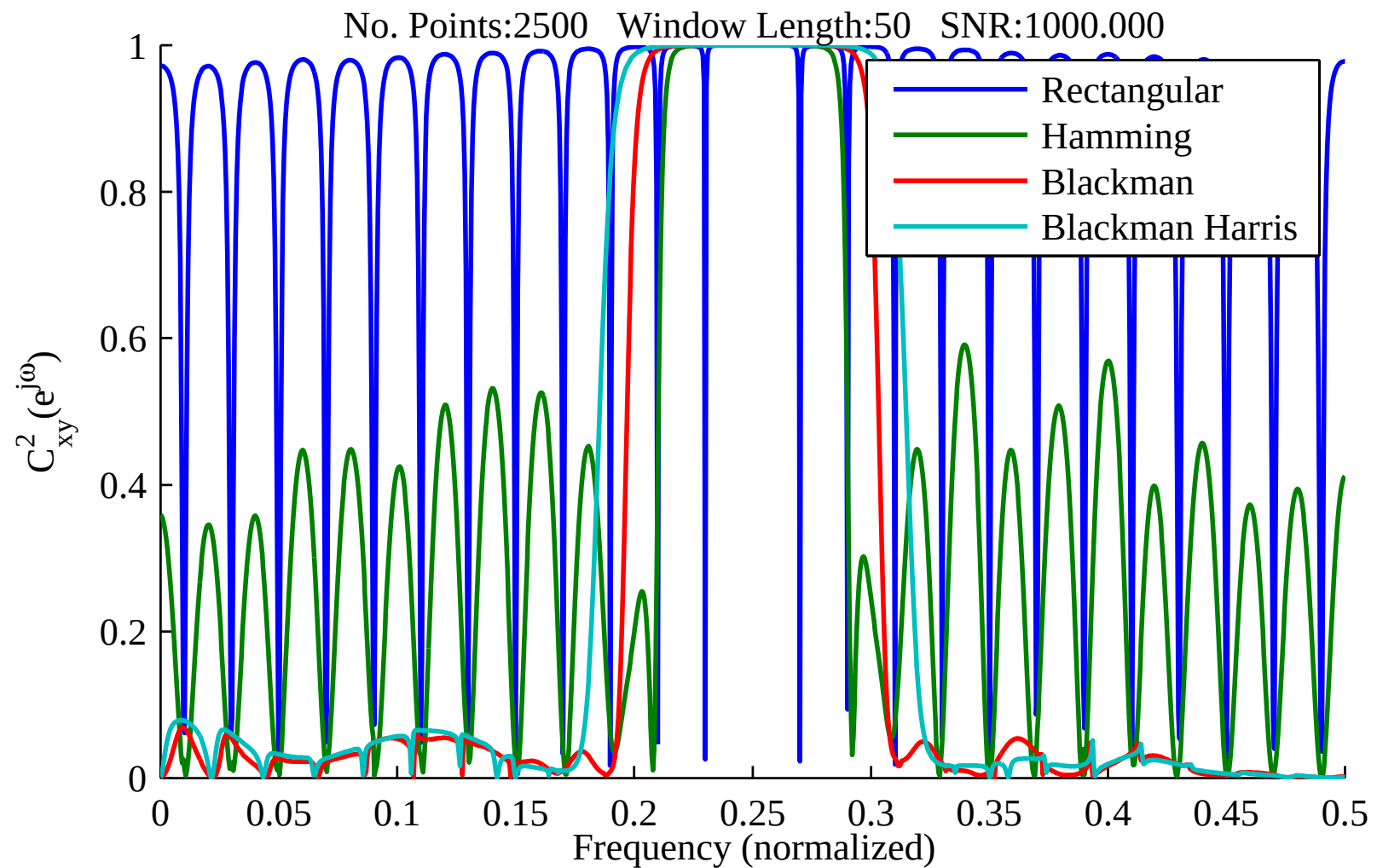
- PSDs with large range (large condition number) are vulnerable to significant spectral leakage
- Preventative measures
 - Pick a data window with very low sidebands
 - Whitening filter (covered next term)
 - Artificial WGN (actually helps!)

Example 9: Spectral Leakage

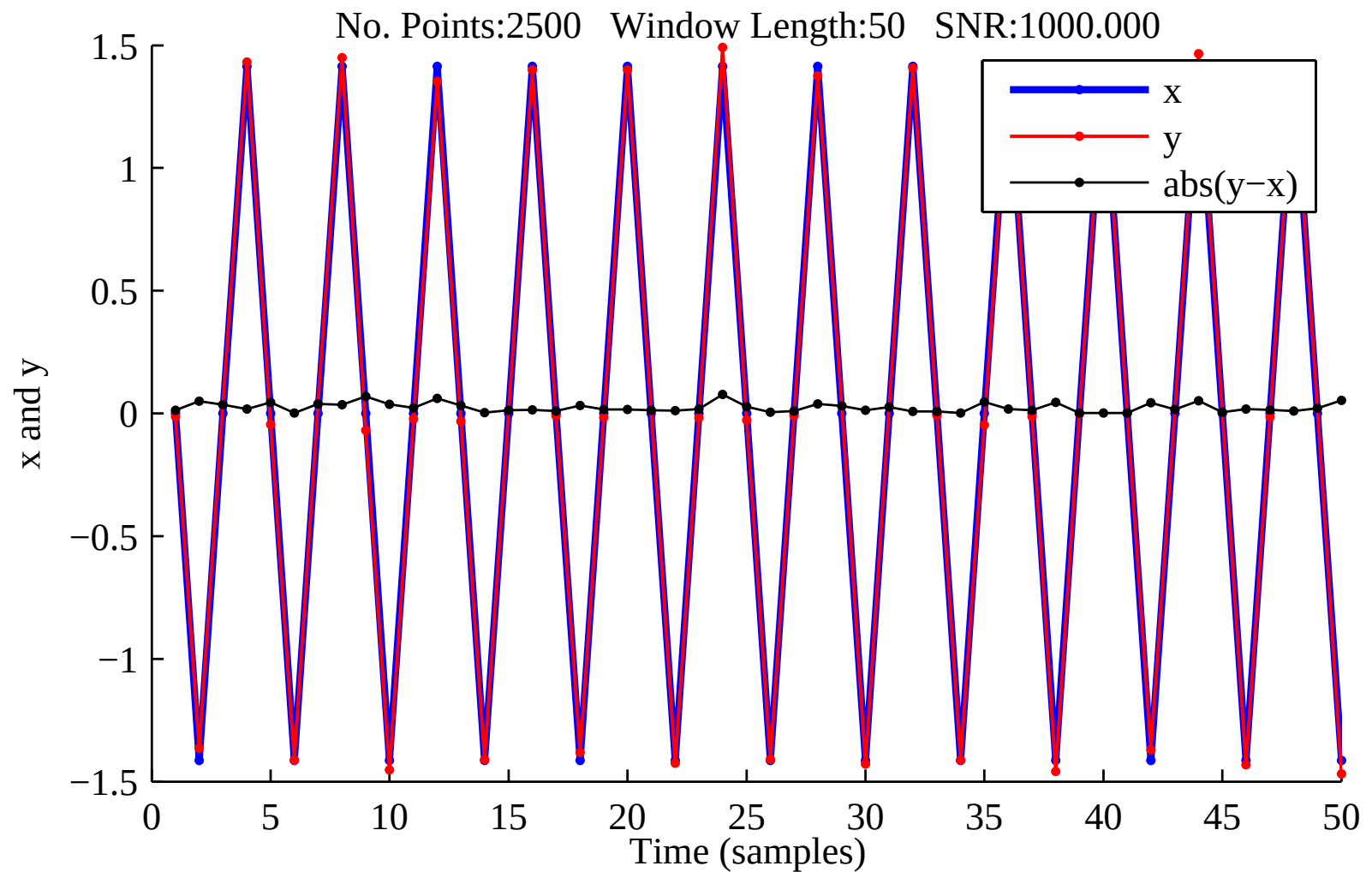
Create a single realization of $N = 1000$ samples of the random process $y(n) = x(n) + \frac{1}{a}w(n)$ where $x(n) = \sqrt{0.5} \cos(\pi/2n)$ and $w(n)$ be zero-mean WGN processes with variance $1/\text{SNR}$. Plot the estimated coherence for a window length of 50 and four different nonparametric windows.

- Which windows have the least amount of bias due to spectral leakage?

Example 9: Spectral Leakage



Example 9: Spectral Leakage



Example 9: MATLAB Code

```
%function [] = Windowing;
close all;
clear all;
wl = 50;

%=====
% Example applied to Coherence
%=====
SNR = 1000;
np = wl*50;
n = (1:np)';
x = sqrt(2)*cos(2*pi*0.25*n);
v = sqrt(1/SNR)*randn(np,1);
y = v + x;

[Crc,f] = Coherency(y,x,1,ones(wl,1),inf,50,2^10);
[Chm,f] = Coherency(y,x,1,hamming(wl),inf,50,2^10);
[Cbm,f] = Coherency(y,x,1,blackman(wl),inf,50,2^10);
[Cbh,f] = Coherency(y,x,1,blackmanharris(wl),inf,50,2^10);

figure;
FigureSet(1,'LTX');
h = plot(f,Crc.^2,f,Chm.^2,f,Cbm.^2,f,Cbh.^2);
set(h,'LineWidth',1.0);
ylim([0 1]);
box off;
xlabel('Frequency (normalized)');
```

```

ylabel('C_{xy}^2(e^{j\omega})');
title(sprintf('No. Points:%d   Window Length:%d   SNR:%5.3f',np,wl,SNR));
AxisSet(8);
legend('Rectangular','Hamming','Blackman','Blackman Harris');
print('WindowsCoherency','-depsc');

figure;
FigureSet(2,'LTX');
k = 1:np;
h = plot(k,x,'b',k,y,'r',k,abs(y-x),'k');
xlim([0 50]);
set(h,'Marker','.');
set(h(2),'LineWidth',0.7);
set(h(1),'LineWidth',1.5);
box off;
xlabel('Time (samples)');
ylabel('x and y');
title(sprintf('No. Points:%d   Window Length:%d   SNR:%5.3f',np,wl,SNR));
AxisSet(8);
legend('x','y','abs(y-x)');
print('WindowsCoherencySignals','-depsc');

return;

%=====
% Window Spectra
%=====
[hrc,w] = freqz(ones(wl,1)/wl,1,2^10);
[hhm,w] = freqz(hamming(wl)/sum(hamming(wl)),1,2^10);
[hbm,w] = freqz(blackman(wl)/sum(blackman(wl)),1,2^10);
[hbh,w] = freqz(blackmanharris(wl)/sum(blackmanharris(wl)),1,2^10);

```

```

figure;
FigureSet(1,'PPT');
f = w/(2*pi);
h = semilogy(f,abs(hrc),f,abs(hhm),f,abs(hbm),f,abs(hbh));
set(h,'LineWidth',1.0);
ylim([1e-6 1]);
box off;
xlabel('Frequency (normalized)');
ylabel('H(e^{j\omega})');
title(sprintf('No. Points:%d',wl));
AxisSet;
legend('Rectangular','Hamming','Blackman','Blackman Harris');
print -dtiff -r150 ../Windows;

%=====
% Example applied to Sinusoid
%=====
n = (1:wl)';
x = 2*cos(2*pi*0.25*n);
Xrc = abs(fft(x.*ones(wl,1)/wl,2^10));
Xhm = abs(fft(x.*hamming(wl)/sum(hamming(wl)),2^10));
Xbm = abs(fft(x.*blackman(wl)/sum(blackman(wl)),2^10));
Xbh = abs(fft(x.*blackmanharris(wl)/sum(blackmanharris(wl)),2^10));
fi = 1:2^9;
f = (fi-1)/2^10;

figure;
FigureSet(2,'PPT');
subplot(2,1,1);
h = plot(f,Xrc(fi),f,Xhm(fi),f,Xbm(fi),f,Xbh(fi));

```

```

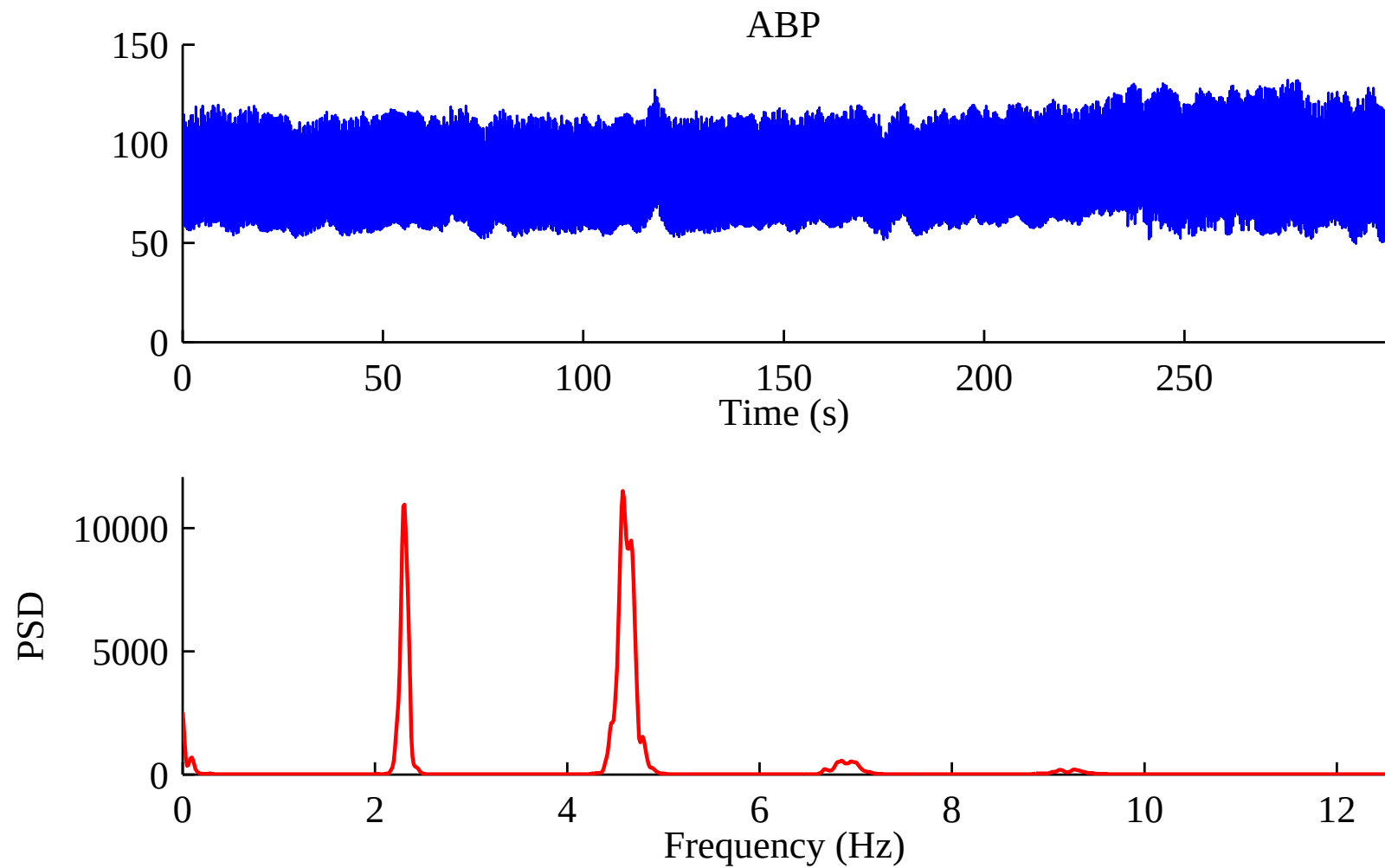
    set(h,'LineWidth',1.0);
    ylim([0 1]);
    box off;
    ylabel('|X(e^{j\omega})|');
    title(sprintf('No. Points:%d',wl));
    AxisSet;
    legend('Rectangular','Hamming','Blackman','Blackman Harris');
subplot(2,1,2);
    h = semilogy(f,Xrc(fi),f,Xhm(fi),f,Xbm(fi),f,Xbh(fi));
    set(h,'LineWidth',1.0);
    ylim([0 1]);
    box off;
    ylabel('|X(e^{j\omega})|');
    xlabel('Frequency (normalized)');
AxisSet;
print -dtiff -r150 ../WindowedSinusoids;

```

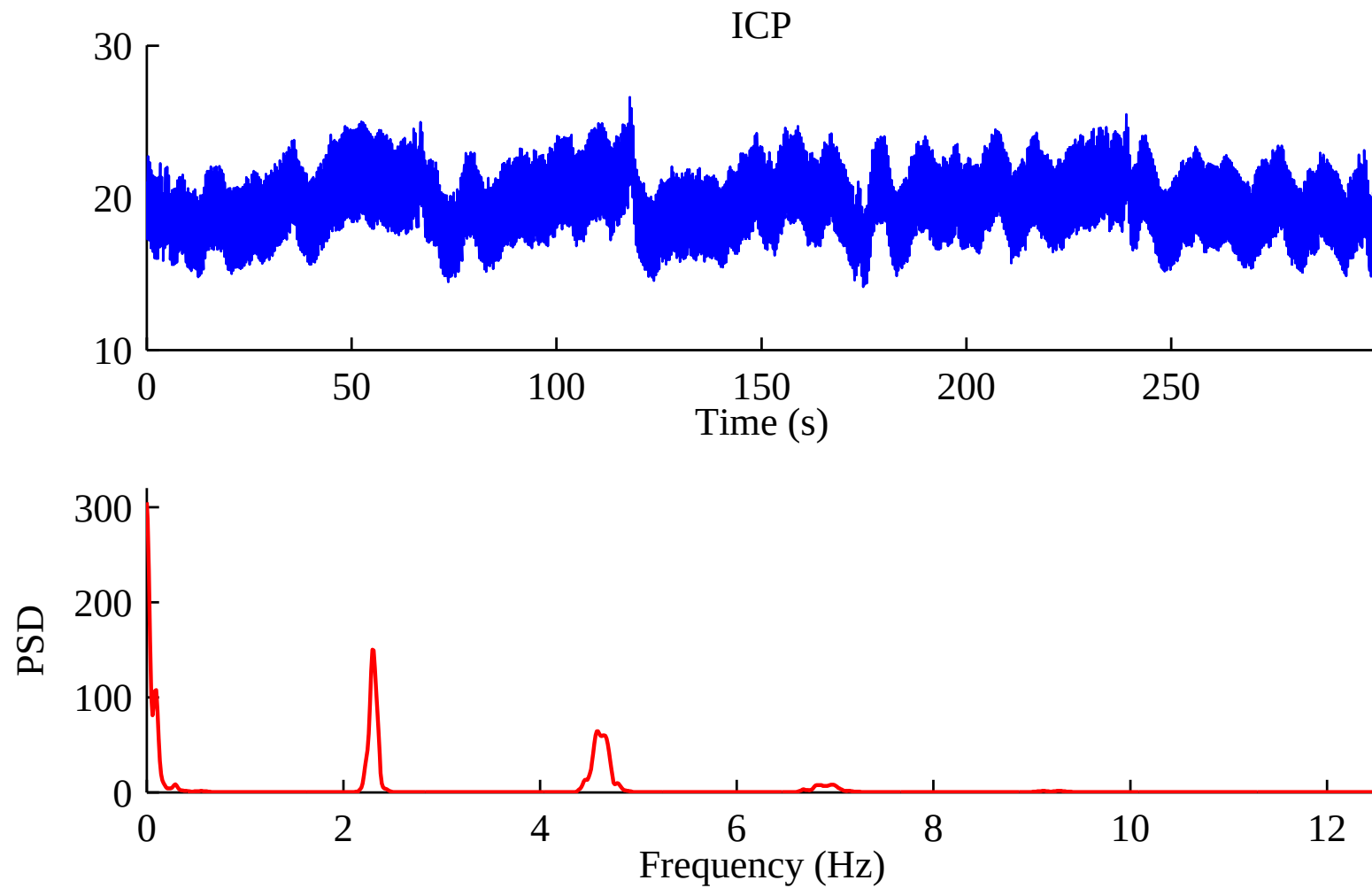
Example 10: Arterial and Intracranial Pressure

Estimate the coherence and PSDs of an intracranial pressure (ICP) and arterial blood pressure (ABP) signals.

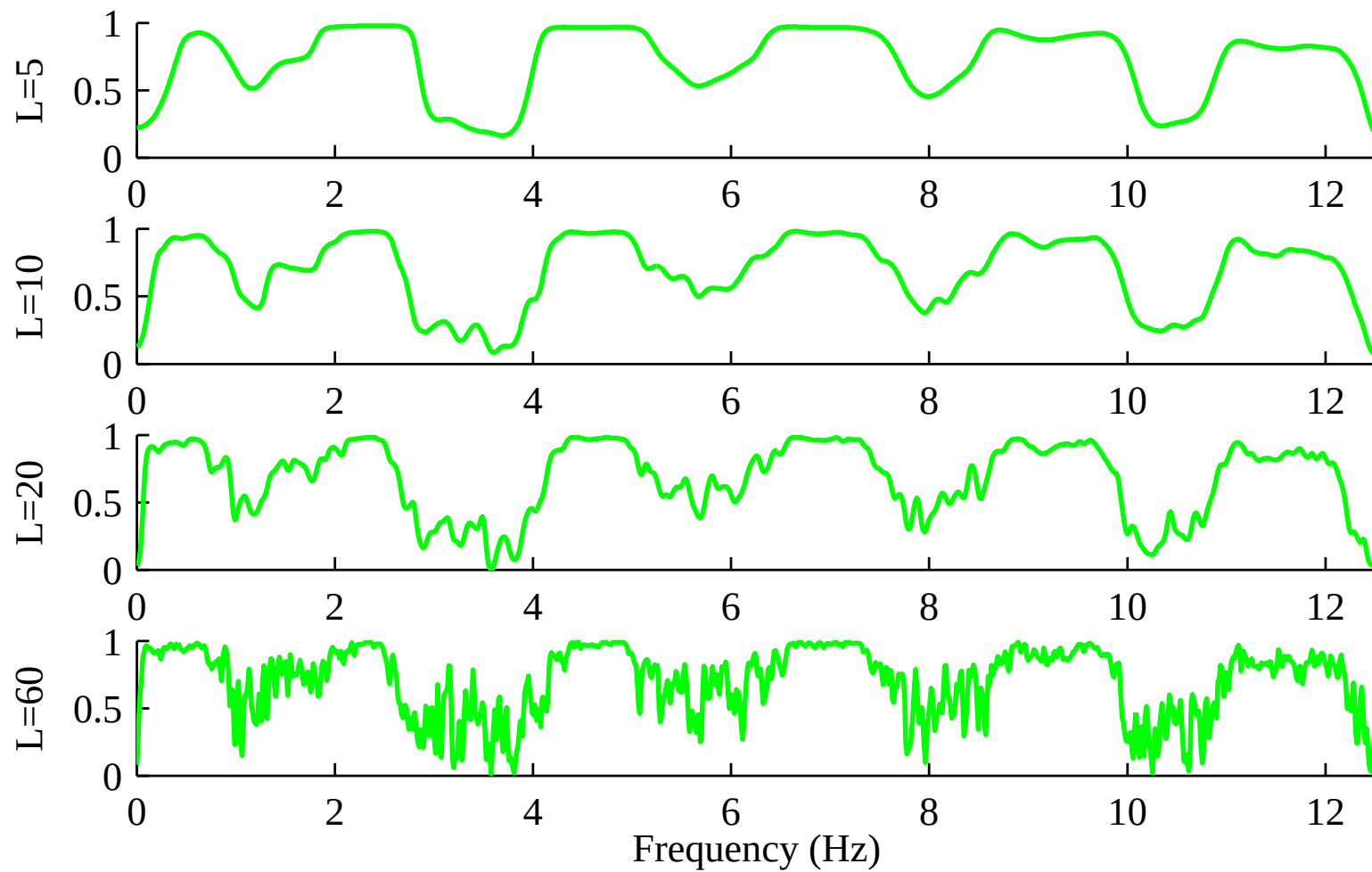
Example 10: ABP PSD



Example 10: ICP PSD



Example 10: ICP-ABP Coherence



Example 10: MATLAB Code

```
clear all;
close all;

load ABPICP.mat;

dr = 5;
wl = 30;
xa = decimate(abp,dr);
xi = decimate(icp,dr);
fs = fs/dr;
xa = xa(5*60*fs + (1:5*60*fs));
xi = xi(5*60*fs + (1:5*60*fs));

for c1=1:2,
    switch c1,
        case 1,
            x = xa;
            ti = 'ABP';
            fn = 'ABPPSD';
        case 2,
            x = xi;
            ti = 'ICP';
            fn = 'ICPPSD';
        end;
    [pa,fa] = Welch(x,fs,wl,50,2^10);
    figure;
```

```

FigureSet(1,'LTX');
subplot(2,1,1);
    k = 1:length(x);
    t = (k-0.5)/fs;
    h = plot(t,x,'b');
    xlim([0 max(t)]);
    xlabel('Time (s)');
    title(ti);
    box off;
subplot(2,1,2);
    h = plot(fa,pa,'r');
    set(h,'LineWidth',0.8);
    xlim([0 fs/2]);
    ylim([0 1.05*max(pa)]);
    xlabel('Frequency (Hz)');
    ylabel('PSD');
    box off;
AxisSet(8);
print(fn,'-depsc');
end;

```

```

figure;
FigureSet(2,'LTX');
wl = [5 10 20 60];
for c1=1:length(wl),
    [c,f] = Coherency(xi,xa,fs,wl(c1),inf,50,2^10);
    subplot(length(wl),1,c1);
    h = plot(f,c.^2,'g');
    set(h,'LineWidth',1.0);
    xlim([0 fs/2]);
    ylim([0 1]);

```

```

        if c1==length(wl),
            xlabel('Frequency (Hz)');
            end;
        ylabel(sprintf('L=%d',wl(c1)));
        box off;
        end;
AxisSet(8);
print('ABPICoherence','-depsc');

figure;
FigureSet(2,'LTX');
wl = [5 10 20 60];
for c1=1:length(wl),
    [c,f] = Coherency(xi,xa,fs,wl(c1),1e3,50,2^10);
    subplot(length(wl),1,c1);
    h = plot(f,c.^2,'g');
    set(h,'LineWidth',1.0);
    xlim([0 fs/2]);
    ylim([0 1]);
    if c1==length(wl),
        xlabel('Frequency (Hz)');
        end;
    ylabel(sprintf('L=%d',wl(c1)));
    box off;
    end;
AxisSet(8);
print('ABPICBiasedCoherence','-depsc');

```

Summary

- Coherence characterizes the degree of linear association as a function of frequency
- Many nice properties
 - Equivalence to the coefficient of determination
 - Natural scale between 0 and 1
 - Unaffected by linear systems
- Difficult to estimate
 - Sensitive to window length and number of segments
 - Approximate confidence intervals
 - Several sources of bias to watch for
- Confidence intervals are wide, even when K and N are large
- Only approximate confidence intervals, since assumptions do not hold in practice

References